Technology Associate Final Round Interview
Case Study

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Final Round Case Study

As part of the Lord Abbett Technology Associate interview process, we ask that you solve one of the below problems and send back to earlycareer@lordabbett.com prior to your Final Round interview:

- Course Schedule
- Price Data Validation
- Fixed Income Analytics Engine

During your Final Round interview, please have your solution pulled up and ready to share with your interviewer. Your interviewer will ask a series of follow-up questions. Please take time in advance to prepare thoroughly. You will be evaluated on your understanding of the problem and relevant data, your ability to design and implement an effective technical solution, the clarity and quality of your code and analysis, and your ability to communicate your reasoning, trade-offs, and recommendations in a real-world business context.

The goal of this exercise is to assess your ability to understand the problem and clearly communicate your reasoning and recommendations.

Good luck!

Course Schedule

Task Description:

There are a total of numCourses courses you have to take, labeled from 0 to *numCourses - 1*. You are given an *array prerequisites* where *prerequisites[i] = [ai, bi]* indicates that you must take course *bi* first if you want to take course ai.

For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return true if you can finish all courses. Otherwise, return false.

Example 1:
Input: numCourses = 2, prerequisites = [[1,0]]
Output: true
Explanation: There are a total of 2 courses to take.
To take course 1 you should have finished course 0. So, it is possible.

Constraints:
- 1 <= numCourses <= 2000
- 0 <= prerequisites.length <= 5000
- prerequisites[i].length == 2
- 0 <= ai, bi < numCourses
- All the pairs prerequisites[i] are unique.

No Program Requirements

Price Data Validation

Task Description:

Implement a Python pipeline to join position and price data, apply fallback logic, compute exposure, and detect outlier prices using z-scores. Use positions.csv, prices.csv, and prices_prev.csv. Output a summary CSV with flags for missing prices, fallback application, exposure, and outliers.

Requirements:

Two data sets:

positions.csv

```
Ticker,Qty
T1,100
T2,200
T3,150
```

prices.csv

```
Ticker,Price,Date
T1,102.33,2025-02-28
T3,2025-02-28
T4,51.2,2025-02-28
```

Outliers: abs(Zscore)>3

Program Requirements: Python, PL/SQL

Fixed Income Analytics Engine

Task Description:

Implement a Yield-to-Maturity solver using Newton-Raphson.

Write a function:

solve_ytm(clean_price, coupon_rate, maturity, settlement, freq, day_count)

You are given 5 bonds:

ID	Type	Coupon	Maturity	Price (clean)	Settlement	Day Count	Freq
B1	UST	3.5%	2032-06-30	98.25	2026-03-15	ACT/ACT	2
B2	IG Corp	5.25%	2029-09-15	101.40	2026-03-15	30/360	2
B3	HY Corp	8.0%	2028-12-01	92.10	2026-03-15	30/360	2
B4	MBS passthrough	4.5%	WAL 6y	99.00	2026-03-15	ACT/360	12
B5	Zero	0%	2027-03-15	95.00	2026-03-15	ACT/ACT	1

Deliverable: Accrued Interest, Dirty Price, YTM

Program Requirements: Python, PL/SQL